Ahmed Amin

AI Researcher | Machine Learning, Deep Learning,NLP and LLMs

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Education

African Institute for Mathematical Sciences (AIMS) Sept 2026 – July 2027 | Cape Town, South Africa

M.Sc. in AI for Science, registered at the **University of Cape Town**
Google DeepMind Scholarship, full funding
Coursework: Bayesian Inference, Applied Machine Learning at Scale, Probabilistic and Scientific Machine Learning, and Quantum Computing

University of Science and Technology at Zewail City Sept 2021 – June 2026 | Giza, Egypt

B.Sc. Communication & Information Engineering (ABET-accredited)
GPA: 3.64/4.00, Cum Laude; Provost's Honors Roll for seven semesters
Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Big Data Analytics, Statistical Inference, Algorithms, Data Structures, and Linear Algebra

Publication

- L. Rana, **A. Amin**, H. Ameer and S. Latif, "FinVerse Agent: Resource-Efficient Financial LLMs with Modular LoRA Fine-Tuning and Task Routing," *2025 IEEE 12th International Conference on Cyber Security and Cloud Computing (CSCloud)*, New York City, NY, USA, 2025, pp. 338–340, doi: 10.1109/CSCloud66326.2025.00061. Adapter code.
- **My contribution (second author):** data preparation and fine-tuning. I prepared and formatted the financial training datasets, then trained task-specific **LoRA** adapters on a 4-bit quantized base model. Only **0.26%** of the model parameters were trained, which kept the cost low. Built with Hugging Face Transformers, PEFT, bitsandbytes and Unsloth.
- The system reached **93.07%** on CFA Level I (GPT-4o: 88.10%) and **97.24 F1** on financial NER (BloombergGPT: 60.82) at **75% lower memory**, and won **1st Place, Task 2, FinAI Contest 2025**. Collaboration between Zewail City and NUST Islamabad.

Research Experience

Research Intern (International Internship) **Jun 2025 – Aug 2025**

National University of Sciences and Technology (NUST), Islamabad, Pakistan

- Selected for an international research internship at NUST's School of Electrical Engineering and Computer Science, starting on site in Islamabad and completing the work remotely.
- Contributed the data preparation and LoRA fine-tuning behind **FinVerse Agent**, the publication listed above.

Professional Experience

AI Engineer (Part-Time) **Dec 2025 – Present**

Takhaial, Giza, Egypt

- Working on efficient local deployment of open-source models: **model quantization**, GPU requirements, and inference optimization for resource-constrained environments.
- Fine-tuned open-source large language models, including **Qwen3-1.7B**, using **Low-Rank Adaptation (LoRA)** and parameter-efficient fine-tuning.
- Built and shipped **four production RAG (retrieval-augmented generation) assistants** to paying clients as part of a small engineering team, contributing across retrieval design, deployment, and system architecture.
- Delivered these assistants on **WhatsApp and the web**, where they run **24/7**, reply in **Arabic and English**, **send images retrieved from the client's database**, **handle voice messages**, and **take live voice calls**.
- Cut per-message inference cost by roughly **5×** by re-engineering the embedding and retrieval layer rather than downgrading the model, preserving answer quality while reducing spend.

Graduation Project

Rafeeq: Smart Wheelchair Companion **2025 – 2026**

Team project | Arabic NLP, Healthcare AI, Edge AI, Robotics

- Won a **$1,600 national grant** from Egypt's Ministry of Higher Education. A retrofit kit pairing an assistive wheelchair with a companion app, at **60–70% below** imported alternatives.
- Built **offline Egyptian Arabic voice control** on a Raspberry Pi 5: **87% accuracy** in **0.9 s**, no internet. Driven by joystick or voice, with LiDAR SLAM navigation (ROS 2 Nav2).
- Built a **RAG chatbot** answering medical questions from an approved knowledge base, with per-patient medication reminders.
- Worked on a **Flutter** app with **separate patient and caregiver views**: live vitals over **Firebase**, chair alerts, and emergency **SOS**.

Selected Projects

- **Dr. Sherine — Arabic retrieval assistant** (details)
  - Built the retrieval layer over **23,512 chunks from 9 classical Arabic books**, bridging a fully diacritized Classical Arabic corpus and undiacritized colloquial Egyptian queries, where surface forms rarely overlap.

- **The Spine — multilingual property sales assistant** (demo, web, WhatsApp)
  - Built the image retrieval and filtering layer over **845 images across 43 buildings** and **192 unit configurations**, resolving floor plans and renders to the specific unit under discussion, with natural-language filtering on price and area across **four languages**.
- **InfinityECN — bilingual trading assistant** (details)
  - Integrated **live market-data APIs** and web search as model tools, with routing that selects between a live tool call and retrieval over the firm's product knowledge base per query. Deployed over web and **telephony voice** in **Gulf Arabic** and English.
- **Movie recommender built from scratch, 32M ratings** (code, details)
  - Derived the alternating least squares equations by hand and wrote them in NumPy with **no recommender library**, training on all **32 million** MovieLens ratings to **0.78 test RMSE** against a 1.06 baseline.
  - Identified compilation rather than parallelism as the bottleneck and obtained a **55×** speed-up with Numba. Trained the final model and wrote up the work as a **paper in the ICML template** (AIMS coursework).
- **Multi-camera threat detection** (code, details)
  - Built a real-time system monitoring concurrent camera feeds for weapons, falls and violence, using a two-stage pipeline where CLIP verifies each detection against text descriptions to suppress false positives.
  - **0.844 mAP@0.50** on weapons and **81%** on violence, running on **CPU only**.
- **Battery scheduling with load forecasting** (code, details)
  - Forecast a building's next-day electricity use, then scheduled a **16 kWh** battery against the real Italian tariff, re-planning at every step instead of trusting one forecast.
  - Within **€0.33** of a perfect-knowledge result and **€10.51** better than the building's own controller.
- **Click-through-rate prediction (D-STN)** (code)
  - Rebuilt the D-STN model (Ouyang et al., KDD 2019) in TensorFlow from the paper alone and reproduced its reported results.
- **Credit approval prediction** (details)
  - The dataset had no label column, so built the target from each client's monthly repayment history and corrected a **29:1** imbalance with SMOTE on the training split only.
  - Best of six models: random forest **96.0%** against a **58.5%** baseline. Coursework, graded A–.
- **Building a language model from scratch** (in progress)
  - Writing a GPT-style model in PyTorch piece by piece on free GPUs: BPE tokenization, RoPE, RMSNorm, SwiGLU, grouped-query attention, supervised fine-tuning, LoRA by hand, and DPO, each part checked against PyTorch's own version.

Technical Skills

- **Programming:** Python, C++, SQL
- **LLMs and Fine-tuning:** LoRA/QLoRA, PEFT, 4-bit quantization, Unsloth, bitsandbytes, Hugging Face Transformers, vLLM
- **RAG and Retrieval:** Hybrid retrieval (BM25 + vector), reranking, LangChain, pgvector, ChromaDB
- **ML and Computer Vision:** PyTorch, TensorFlow, scikit-learn, XGBoost, NumPy, Numba, CLIP, Whisper
- **Data and Backend:** PostgreSQL, MongoDB, Redis, FastAPI, REST APIs, microservices, Celery
- **Infrastructure:** AWS EC2, Docker, Docker Swarm, Linux, Git, CI/CD, Promtail/Loki

Scholarships & Awards

- **Google DeepMind Scholarship**, full funding for the AI for Science M.Sc. 2026
- **1st Place, FinAI Contest 2025**, Task 2 2025
- **Ministry of Higher Education Grant**, Innovators Support Fund, for Rafeeq 2026
- **Full Excellence Scholarship**, Zewail City 2021–2026
- **Zewail City Recognition Awards**, Entrepreneurship, Scientific Research, and Community Service 2026

Leadership & Teaching

Egypt Scholars, Basic Lab 12 **Feb 2026 – May 2026**
Competitively selected
- Mentored peers and trained in research methods and scientific writing.

Elected Student Representative, Zewail City **2021 – 2025**
- Elected four years running to represent all four CIE classes, serving **200+ students**.

Coding Tutor, DEMI Program, iSchool **Jun 2023 – Aug 2023**
- Taught programming to children in grades 4–6.

Teaching Assistant, Calculus I, Zewail City **Oct 2022 – Jan 2023**
- Ran tutorials and one-to-one help for **50+ first-year students**.

Languages

Arabic: Native (MSA and Egyptian dialect) **English:** Professional proficiency (IELTS Academic 6.5, CEFR B2)